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Time taken	50 secs
Grade	80.00 out of 100.00

Question 1

Correct

Mark 20.00 out of 20.00

Write a python program to implement linear search on the given tuple of string values.

note: As the tuple is immutable convert the list to tuple to perform search

For example:

Input	Result
5 ram john akbar seetha oviya john	Tuple: john found
4 rohini fathima jenifer nizam rakesh	Tuple: rakesh not found

Answer: (penalty regime: 0 %)

```

1 def LS(list,n,key):
2     for i in range(0,n):
3         if(list[i]==key):
4             return i
5     return -1
6 list=[]
7 n=int(input())
8 for i in range(0,n):
9     temp=input()
10    list.append(temp)
11 key=input()
12 res=LS(list,n,key)
13 if(res==-1):
14     print("Tuple: %s not found"%key)
15 else:
16     print("Tuple: %s found"%key)

```

	Input	Expected	Got	
✓	5 ram john akbar seetha oviya john	Tuple: john found	Tuple: john found	✓
✓	4 rohini fathima jenifer nizam rakesh	Tuple: rakesh not found	Tuple: rakesh not found	✓

	Input	Expected	Got	
✓	6 rose jasmine tulips marigold hibiscus lotus lilly	Tuple: lilly not found	Tuple: lilly not found	✓

Passed all tests! ✓



Marks for this submission: 20.00/20.00.

Question 2

Correct

Mark 20.00 out of 20.00

Write a python program to implement quick sort on the given float array values.

For example:

Input	Result
5	left: []
6.9	right: []
8.3	left: []
2.1	right: []
1.5	left: [1.5]
6.4	right: [6.4]
	left: []
	right: []
	left: [1.5, 2.1, 6.4]
	right: [8.3]
	[1.5, 2.1, 6.4, 6.9, 8.3]
6	left: []
3.1	right: []
2.4	left: []
5.6	right: []
4.3	left: []
6.2	right: []
7.8	left: []
	right: [7.8]
	left: [4.3]
	right: [6.2, 7.8]
	left: [2.4]
	right: [4.3, 5.6, 6.2, 7.8]
	[2.4, 3.1, 4.3, 5.6, 6.2, 7.8]

Answer: (penalty regime: 0 %)

```

1 def qsort(L):
2     if L==[]:
3         return[]
4     pivot=L[0:1]
5     left=qsort([x for x in L[1:]if x<L[0]])
6     right=qsort([x for x in L[1:]if x>=L[0]])
7     print("left: ",left)
8     print("right: ",right)
9     return left+pivot+right
10 li=[]
11 n=int(input())
12 for i in range(n):
13     li.append(float(input()))
14 print(qsort(li))

```

	Input	Expected	Got	
✓	5 6.9 8.3 2.1 1.5 6.4	left: [] right: [] left: [] right: [] left: [1.5] right: [6.4] left: [] right: [] left: [1.5, 2.1, 6.4] right: [8.3] [1.5, 2.1, 6.4, 6.9, 8.3]	left: [] right: [] left: [] right: [] left: [1.5] right: [6.4] left: [] right: [] left: [1.5, 2.1, 6.4] right: [8.3] [1.5, 2.1, 6.4, 6.9, 8.3]	✓
✓	6 3.1 2.4 5.6 4.3 6.2 7.8	left: [] right: [] left: [] right: [] left: [] right: [] left: [] right: [7.8] left: [4.3] right: [6.2, 7.8] left: [2.4] right: [4.3, 5.6, 6.2, 7.8] [2.4, 3.1, 4.3, 5.6, 6.2, 7.8]	left: [] right: [] left: [] right: [] left: [] right: [] left: [] right: [7.8] left: [4.3] right: [6.2, 7.8] left: [2.4] right: [4.3, 5.6, 6.2, 7.8] [2.4, 3.1, 4.3, 5.6, 6.2, 7.8]	✓
✓	8 1.2 1.3 4.2 5.3 6.4 7.3 6.8 9.2	left: [] right: [] left: [] right: [] left: [6.8] right: [9.2] left: [] right: [6.8, 7.3, 9.2] left: [] right: [6.4, 6.8, 7.3, 9.2] left: [] right: [5.3, 6.4, 6.8, 7.3, 9.2] left: [] right: [4.2, 5.3, 6.4, 6.8, 7.3, 9.2] left: [] right: [1.3, 4.2, 5.3, 6.4, 6.8, 7.3, 9.2] [1.2, 1.3, 4.2, 5.3, 6.4, 6.8, 7.3, 9.2]	left: [] right: [] left: [] right: [] left: [6.8] right: [9.2] left: [] right: [6.8, 7.3, 9.2] left: [] right: [6.4, 6.8, 7.3, 9.2] left: [] right: [5.3, 6.4, 6.8, 7.3, 9.2] left: [] right: [4.2, 5.3, 6.4, 6.8, 7.3, 9.2] left: [] right: [1.3, 4.2, 5.3, 6.4, 6.8, 7.3, 9.2] [1.2, 1.3, 4.2, 5.3, 6.4, 6.8, 7.3, 9.2]	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 3

Correct

Mark 20.00 out of 20.00

Write a python program to implement merge sort using iterative approach on the given list of float values.

For example:

Test	Input	Result
Merge_Sort(S)	5 10.2 21.3 3.5 7.8 9.8	The Original array is: [10.2, 21.3, 3.5, 7.8, 9.8] Array after sorting is: [3.5, 7.8, 9.8, 10.2, 21.3]
Merge_Sort(S)	6 20.3 41.2 5.3 6.2 8.1 65.2	The Original array is: [20.3, 41.2, 5.3, 6.2, 8.1, 65.2] Array after sorting is: [5.3, 6.2, 8.1, 20.3, 41.2, 65.2]

Answer: (penalty regime: 0 %)

```

1 def merge(S, temp, From, mid, to):
2     a = From
3     b = From
4     c = mid + 1
5     while b <= mid and c <= to:
6         if S[b] < S[c]:
7             temp[a] = S[b]
8             b = b + 1
9         else:
10            temp[a] = S[c]
11            c = c + 1
12            a = a + 1
13     while b < len(S) and b <= mid:
14         temp[a] = S[b]
15         a = a + 1
16         b = b + 1
17     for b in range(From, to + 1):
18         S[b] = temp[b]
19 def Merge_Sort(S):
20     low = 0
21     high = len(S) - 1
22     temp = S.copy()

```

	Test	Input	Expected	Got	
✓	Merge_Sort(S)	5 10.2 21.3 3.5 7.8 9.8	The Original array is: [10.2, 21.3, 3.5, 7.8, 9.8] Array after sorting is: [3.5, 7.8, 9.8, 10.2, 21.3]	The Original array is: [10.2, 21.3, 3.5, 7.8, 9.8] Array after sorting is: [3.5, 7.8, 9.8, 10.2, 21.3]	✓
✓	Merge_Sort(S)	6 20.3 41.2 5.3 6.2 8.1 65.2	The Original array is: [20.3, 41.2, 5.3, 6.2, 8.1, 65.2] Array after sorting is: [5.3, 6.2, 8.1, 20.3, 41.2, 65.2]	The Original array is: [20.3, 41.2, 5.3, 6.2, 8.1, 65.2] Array after sorting is: [5.3, 6.2, 8.1, 20.3, 41.2, 65.2]	✓

	Test	Input	Expected	Got	
✓	Merge_Sort(S)	4 2.3 6.1 4.5 96.5	The Original array is: [2.3, 6.1, 4.5, 96.5] Array after sorting is: [2.3, 4.5, 6.1, 96.5]	The Original array is: [2.3, 6.1, 4.5, 96.5] Array after sorting is: [2.3, 4.5, 6.1, 96.5]	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **4**

Correct

Mark 20.00 out of 20.00

Write a Python Program Using a recursive function to calculate the sum of a sequence**For example:**

Input	Result
20	210
36	666
45	1035

Answer: (penalty regime: 0 %)

```

1 def SumOf(n):
2     if n>0:
3         return n+SumOf(n-1)
4     return 0
5 n = int(input());
6 res = SumOf(n);
7 print(res);

```

	Input	Expected	Got	
✓	20	210	210	✓
✓	36	666	666	✓
✓	45	1035	1035	✓
✓	58	1711	1711	✓
✓	65	2145	2145	✓

Passed all tests! ✓



Marks for this submission: 20.00/20.00.

Question 5

Not answered

Mark 0.00 out of 20.00

Write a python program to implement linear search on the given tuple of float values.

note: As the tuple is immutable convert the list to tuple to perform search

For example:

Input	Result
5 3.2 1.5 6.4 7.8 9.5 6.4	Tuple: 6.4 found
6 3.2 1.2 3.4 5.3 6.2 6.8 6.2	Tuple: 6.2 found

Answer: (penalty regime: 0 %)

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